Finite Families of Subspaces of Codimension at Least Two Are Never Helmholtz Uniqueness Sets

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Source Question and Status

Fernández-Bertolin, Gröchenig, and Jaming prove nonuniqueness for an arbitrary finite collection of lines in $\mathbb{R}^d$. Immediately afterwards they write: "We do not know what happens for restrictions of the Helmholtz equation to k-dimensional subspaces when $2 \le k \le d-2$ and $d \ge 4$" [1, p. 18].

> We do not know what happens for restrictions of the Helmholtz equation to k-dimensional subspaces when $2 \le k \le d-2$ and $d \ge 4$.

In the fixed finite-family setting of their preceding general problem, the answer is always negative. In fact, no finite family of linear subspaces of codimension at least two is a uniqueness set, for any real wave number. The same construction disproves the associated sphere/subspace Heisenberg uniqueness property. Status: *candidate full counterexample theorem, likely valid*, subject to human review.

Proof Intuition

Vanishing on a whole k-plane imposes more than finitely many point conditions, but at a fixed homogeneous degree it still imposes only $O(m^{k-1})$ linear conditions. Degree-m spherical harmonics in d variables have dimension $O(m^{d-2})$. When $k \le d-2$, a finite number of restriction maps therefore cannot detect every high-degree spherical harmonic. A common kernel harmonic gives a harmonic counterexample directly. Multiplying it by the regular radial Helmholtz power series gives the same counterexample at every wave number, without a singularity at the origin.

Main Result

Let $\mathcal{H}_m^d$ be the real vector space of homogeneous harmonic polynomials of degree m on $\mathbb{R}^d$, and let $\mathcal{P}_m(E)$ be the homogeneous degree-m polynomials on a linear space E.

Theorem 1 (finite subspaces are not unique)**.** *Let $d \ge 3$, let $E_1, \dots, E_N \subset \mathbb{R}^d$ be linear subspaces with $1 \le k_j := \dim E_j \le d-2$, and let $\lambda \in \mathbb{R}$. There exists a nonzero real entire function u on $\mathbb{R}^d$ such that*

$$(\Delta + \lambda^2)u = 0, \qquad u|_{E_1 \cup \cdots \cup E_N} = 0.$$

Consequently, for $d \ge 4$ and $2 \le k \le d-2$, no finite collection of k-dimensional linear subspaces is a uniqueness set for Helmholtz solutions.

Proof. For each $m \geq 1$, consider the direct-sum restriction map

$$\mathcal{R}_m : \mathcal{H}_m^d \longrightarrow \bigoplus_{j=1}^{N} \mathcal{P}_m(E_j), \qquad H \longmapsto (H|_{E_1}, \dots, H|_{E_N}).$$

The Fischer decomposition of homogeneous polynomials gives

$$\begin{aligned} \dim \mathcal{H}_m^d &= \binom{m+d-1}{d-1} - \binom{m+d-3}{d-1} \\ &= \binom{m+d-2}{d-2} + \binom{m+d-3}{d-2} = \frac{2}{(d-2)!} m^{d-2} + O(m^{d-3}). \end{aligned} \tag{1}$$

On the other hand,

$$\dim \mathcal{P}_m(E_j) = \binom{m+k_j-1}{k_j-1} = O(m^{k_j-1}) = O(m^{d-3}). \tag{2}$$

Since N is fixed, equations (1)–(2) show that for all sufficiently large m,

$$\dim \mathcal{H}_m^d > \sum_{j=1}^{N} \dim \mathcal{P}_m(E_j).$$

Hence $\ker \mathcal{R}_m$ contains a nonzero real polynomial H. Thus $H|_{E_j} = 0$ for every j.

Put $s = |x|^2$ and define the entire scalar function

$$\Phi_m(s) = \sum_{\ell=0}^{\infty} \frac{(-\lambda^2 s/4)^\ell}{\ell!\,(m+d/2)_\ell}, \qquad u(x) = H(x)\Phi_m(|x|^2), \tag{3}$$

where $(a)_\ell$ is the rising factorial. The series has infinite radius of convergence and $\Phi_m(0) = 1$, so u is entire and nonzero. Since H is homogeneous of degree m and harmonic,

$$\Delta\big(H(x)\Phi(|x|^2)\big) = H(x)\{4s\Phi''(s) + (4m+2d)\Phi'(s)\}. \tag{4}$$

If a_ℓ denotes the coefficient of s^ℓ in (3), then

$$4(\ell+1)(\ell+m+d/2)a_{\ell+1} + \lambda^2 a_\ell = 0.$$

Substitution in (4) gives $(\Delta + \lambda^2)u = 0$. Finally, the factor H shows that u vanishes identically on every E_j. $\square$

Heisenberg Uniqueness Consequence

Corollary 1. *Under the assumptions of Theorem 1, $(\mathbb{S}^{d-1}, E_1 \cup \cdots \cup E_N)$ is not a Heisenberg uniqueness pair. More strongly, there are two distinct finite positive measures on $\mathbb{S}^{d-1}$ whose Fourier transforms agree on every E_j.*

Proof. Choose $m \geq 1$ and $0 \neq H \in \ker \mathcal{R}_m$ as above, and let $Y = H|_{\mathbb{S}^{d-1}}$. After scaling, the measures

$$d\mu_\pm(\theta) = \left(1 \pm \frac{Y(\theta)}{\|Y\|_\infty}\right) d\sigma(\theta)$$

are positive and distinct. The Funk–Hecke formula gives, for a nonzero constant $c_{m,d}$ and $r > 0$,

$$\widehat{Y\sigma}(r\omega) = c_{m,d}\, r^{-(d-2)/2} J_{m+(d-2)/2}(r) Y(\omega).$$

If $r\omega \in E_j$, then $Y(\omega) = 0$. At the origin the transform also vanishes because every positive-degree spherical harmonic has mean zero. Thus $\widehat{\mu_+} = \widehat{\mu_-}$ on each E_j. $\square$

Scope, Novelty Check, and Verification

The result answers the source's stated intermediate-dimensional linear- subspace case for every finite configuration. It does not claim nonuniqueness for infinitely many subspaces or for arbitrary curved lower-dimensional manifolds. The codimension threshold is sharp for this argument: when $\dim E = d - 1$, the source proves uniqueness for two generic intersecting hypersurfaces, and the domain and restriction dimensions have the same polynomial order.

The run's cheap indexes contained no answer. Exact-sentence and concept searches through 17 August 2026 found the source, its published version, and related HUP/PDE literature, but no later answer to this finite-family question. The companion script checks the exact dimension identities, finds kernel-producing degrees across a grid of dimensions and family sizes, and verifies the radial coefficient recurrence using rational arithmetic. These are consistency checks; the proof above is exact.

Human Review Recommendation

Check that the source's open sentence inherits the finite-family setting from the preceding general problem, and verify the rank bound $\operatorname{rank}(H \mapsto H|_E) \leq \dim \mathcal{P}_m(E)$. Once those two points are accepted, the dimension gap and the radial recurrence settle the linear-subspace problem.

References

[1] A. Fernández-Bertolin, K. Gröchenig, and P. Jaming, *From Heisenberg uniqueness pairs to properties of the Helmholtz and Laplace equations*, Journal of Mathematical Analysis and Applications **469** (2019), 202–219; arXiv:1711.05520; doi:10.1016/j.jmaa.2018.09.008.